

A Review Paper on Question Classification and Automated Question Generation Techniques Based on Bloom's Taxonomy



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Abstract In today's education system, it is important to have fast and efficient algorithms for categorizing and generating exam questions across cognitive levels. Automating this process aids in assessments that align with learning objectives, improving both evaluation and learning outcomes. Bloom's taxonomy, a framework that classifies learning objectives into different levels of complexity and understanding, is widely used in modern pedagogy by promoting higher-order learning and critical thinking. This paper presents a comprehensive review of existing approaches for both question classification and automated question generation, making it the first study to analyze these two aspects together. It focuses on methods ranging from traditional machine learning algorithms to advanced deep learning techniques, evaluating their effectiveness in improving efficiency and accuracy across both domains. With the growing interest in taxonomy-based learning approaches, this research is very useful in offering insights into how AI-driven techniques can improve question classification and automated question generation. The main finding of this review is identifying research gaps, particularly the absence of practical application of these methods in real-world educational tools. There is increasing interest in designing syllabus based on Bloom's taxonomy. By bridging this gap, future research can focus on developing intelligent educational platforms that dynamically generate and classify questions to enhance student learning. The aim of this review is to provide an overview of recent approaches and techniques proposed in question classification and automated question generation. It provides valuable insights for researchers and educators aiming to implement efficient and scalable techniques in real-world educational environments, eventually improving the design and implementation of question generation and question classification tools.

Keywords Bloom's taxonomy · Question classification · Automatic question generation · NLP · Transformer

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1 Introduction

In today's ever-evolving educational field, the expertise to assess students' cognitive skills effectively plays a crucial role for nurturing higher-order learning and critical thinking. Two important aspects of modern assessment include ensuring cognitive-level-based evaluation and continuous new question generation to meet the demands. For this purpose, a widely used framework called Bloom's taxonomy (BT) was introduced by Benjamin Bloom in 1956, as shown in Fig. 1. BT advances through six cognitive levels, beginning with remember, where students recollect facts; understand, where they interpret meaning; apply, where they utilize the knowledge of previous two levels; analyze, where they break down and relate the information; evaluate, where they make judgments based on criteria; and then create, where they construct ideas into new forms implying higher-order thinking skills. Classification and generation of new questions based on BT ensure that assessment covers a balanced mix of cognitive levels, enabling educators to evaluate a student's mastery across cognitive levels. However, manual generation of voluminous questions and their classification can be a tedious task requiring more manual effort and time. As educational institutions continuously rely on today's technology to manage plentiful assessments, this paper highlights the need for automated systems that can accurately classify and generate exam questions based on Bloom's cognitive levels that reflect evolving curricula. Today's educational era demands to develop systems that not only analyzes the language of the questions but also understands the context in which the questions are framed. This paper reviews various approaches, ranging from traditional machine learning [1, 3, 10–12, 19–21, 25] to advanced deep learning [4, 8, 15, 22], which have been used previously for the automated classification and question generations [9, 17, 18, 24] based on BT. The primary problem this review deals with is: "What are the different methods previously used to classify exam questions and to generate new questions based on Bloom's cognitive levels?" This question is critical as it tells us the effect of previously used techniques and their findings which can be used for future developments. The core finding of this review is that while

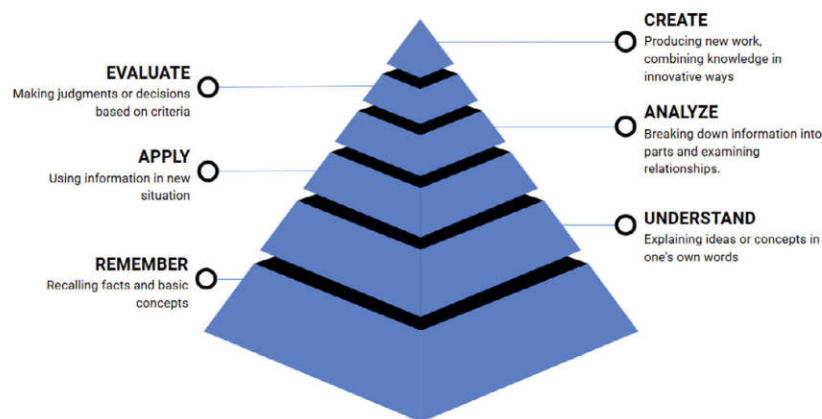


Fig. 1 Revised Bloom's taxonomy

various approaches have been proposed for the automated question classification and generation task, the combination of these approaches provides desired accuracy, scalability, and enhanced performance. Owing to this, there is need to improve and innovate in this field to fulfill demands of this ever-evolving education field. Our motivation for this review stems from the increasing need for automation in educational systems. By understanding the gaps in current research, we aim to contribute to the development of tools that not only simplify the question classification process but also ensure a personalized learning experience for students. An implied way for students to prepare for exams is by practicing with existing question papers. However, it is important to make sure that the preparation includes questions belonging to all cognitive levels defined by BT. Analyzing which cognitive levels are more frequently tested in exams can further help students prioritize their preparation, enabling more focused and effective studying. Additionally, we aim to lay the groundwork for the development of a platform that classifies past year questions (PYQs) and modify the question-forming process that aligns with the student's cognitive development needs.

2 Literature Survey

2.1 *Classification Based on Bloom's Taxonomy*

In recent years, researchers have used many approaches (Refer Table 1) to classify questions related to exams based on BT.

1. Rule-based approach

Rule-based approaches use predefined rules based on syntactic features to classify questions according to the cognitive levels of BT. Harris et al. [10] examined rule-based approach, N-grams, and a combination of both methods to categorize questions by utilizing BT. The dataset involved 135 programming questions divided into 100 and 35 questions for training and testing, respectively. Jayakodi et al. [12] introduced a new set of rules that use NLP and WordNet similarity algorithms along with a cosine similarity algorithm to classify questions according to BT.

2. Machine learning approach

Over recent years, various machine learning approaches have been used for classification of questions across different cognitive levels based on BT. Yahya et al. [25] examined the performance of support vector machine (SVM) using an item bank questions dataset consisting of 600 questions. Abduljabbar et al. [1] introduced an ensemble of SVM, NB, K-NN classifiers combined using voting algorithms and feature selection methods, namely Chi-square, Mutual Information, and Odd Ratio. The model trained and tested on the exam question bank of programming questions resulted in better performance than individual classifiers. Aninditya et al. [3] used a collection of exam questions from the mid-term and final exams

Table 1 Comparison between various research studies in classification

Author	Approach	Technique	Outcome	Gap analysis
[10]	Rule-based	Rules using NLP analyzed syntactic structures to identify keywords and verbs, while N-grams were used when rules failed to categorize questions	The hybrid technique had an 86% F1-score, making it 1.34 times better than the rule-based method and 1.7 times better than N-grams, showing that combining both worked best	Limited dataset, time-consuming and expensive technique, less robust compared to machine learning techniques
[12]	Rule-based	WordNet and cosine similarity were used to create rule sets and assign weights for classifying questions based on BT	The classification process's result has 71% accuracy rate	As the research utilized essay-type questions, MCQ categorization was not explored
[25]	ML-based	Conducted two experiments: one to evaluate SVM performance with different term frequency (TF) values and another to examine the impact of stopword removal	SVM performance declines as the number of terms representing a question increases, with the best results achieved when $TF \geq 2$. Also, removing stopwords does not significantly improve performance	Did not explore ensemble learning techniques and other classifiers
[1]	ML-based	The study had two phases: first, an ensemble of SVM, NB, and K-NN classifiers; second, a voting-based combination strategy using feature selection methods like Chi-square, Mutual Information, and Odds Ratio	The use of feature selection methods improved the results of classifiers with height result among feature selection methods were obtained when applying Mutual Information at feature size 250	Did not explore DL approaches

(continued)

Table 1 (continued)

Author	Approach	Technique	Outcome	Gap analysis
[3]	ML-based	Utilized NB and TF-IDF with N-grams	Classification using NB and TF-IDF with N-grams as indexing term achieved precision of 85% and recall of 80%	Lacks investigation into other classifiers
[11]	ML-based	Implemented and compared multiple ML techniques such as K-Nearest Neighbors, Random Forest, Decision Trees, SVM, Neural Network, Linear Discriminant Analysis (LDA), and Logistic Regression	Best result with Logistic Regression and Linear Discriminant Analysis (LDA) Machine Learning techniques both having an accuracy of 83.3%	– Insufficient instances (exceptions in dataset) – Missing some action verbs and ignoring question length as a feature
[21]	ML-based	A multi-agent system mapped questions to Bloom's levels. NLP techniques processed students' responses, using SVM to predict question levels and Random Forest to evaluate answers	Highest accuracy: SVM (98%) and Random Forest (92%) Subjective question handling	– Limited subject coverage – Agents are predefined, not dynamically trained
[15]	DL-based	A model is proposed in this research work for the classification of question items based on both cognitive process and knowledge dimensions by using CNN and LSTM	CNN achieved 80% accuracy, outperforming LSTM in testing (66.67% vs. 44%). Results help identify student weaknesses and inform question paper design	– Limited Dataset Size – Focus on a Single Subject – Manual Labeling Bias

(continued)

Table 1 (continued)

Author	Approach	Technique	Outcome	Gap analysis
[22]	DL-based	Created a pipeline that starts by generating questions, then classifying them and then providing explicit recommendations for context-based questions	BERT-based embeddings with a two-layer dense NN using Adam optimizer with a slow learning rate of $1e-5$. Model achieved 81.1% validation accuracy, 0.86 validation loss, and an F1-score of 0.81	– Limited computational resources for advanced architectures – A small SQuAD dataset used (3000 entries) – Focused only on original BT, requiring further exploration
[8]	DL-based	The study compared 6 embedding techniques (3 non-contextual, 3 contextual). It used a CNN model with RoBERTa for classifying exam questions by BT cognitive levels, outperforming previous models	RoBERTa is the best embedding for classifying questions based on BT. Introduced a CNN model that outperformed previous models with weighted F1-scores and 86% accuracy	- Limited dataset size needs evaluation on larger datasets - Accuracy can improve with hybrid architectures like CNN-RNN - Fine-tuning large models like BERT and RoBERTa is computationally complex compared to CNN
[4]	DL-based	A NCERT-based dataset improved BT question classification. ML/DL models outperformed rule-based methods, and fine-tuning on a combined dataset boosted performance by 3%. Weighted Average Ensemble improved prediction accuracy	Achieved 94.10% accuracy with the proposed dataset, beating previous models (92.77%, 88.50%, 78.44%). Fine-tuning improved accuracy by 3%. The best ensemble model (ALBERT + DeBERTa + ELECTRA) outperformed traditional ML models	– Incomplete coverage of question types across subjects. Only supports text-based questions, no images/diagrams – Inconsistent cognitive level labeling due to human annotation

of a university and utilized NB and TF-IDF along with N-grams. Jain et al. [11] performed and compared multiple ML techniques such as K-nearest neighbors, random forest, decision trees, SVM, neural network, linear discriminant analysis (LDA), and logistic regression. Shahzad et al. [21] used web scraped software engineering questions and classifying them using SVM classifier. She further created six different agents for each level in BT to handle them independently.

3. Deep learning approach

Manjushree et al. [15] experimented with LSTM and CNN models for classification. The models were compared based on their results on the dataset that consisted of 844 software engineering questions from different institutions. Sharma et al. [22] included BT-based classification inside a pipeline that included question generation and recommendation for context-based questions. The dataset they used was SQuAD 2.0 which is a reading comprehension dataset that has questions and answers based on Wikipedia as context. Along with which they also surveyed faculty members to get manually annotated questions from passages in school textbooks. Gani et al. [8] used two datasets that were a combination of five datasets. The study was done in two phases. The first phase was dedicated to find the optimal word embedding technique which concluded that RoBERTa was the best technique. The second phase was about hyperparameter tuning and achieved precision, recall, F1-score of 0.85, 0.72, and 0.78, respectively. Chindukuri et al. [4] used an ensemble of pre-trained models, namely BERT, RoBERTa, DeBERTa, ELECTRA, ALBERT on 7064 questions from NCERT textbooks of grades VI to XII.

2.2 Automatic Question Generation (AQG) Based on Bloom's Taxonomy

Bambang et al. [24] designed a model to generate educational questions aligned with Bloom's taxonomy by extracting key phrases and applying context-free grammar rules. Using software engineering course material for testing, the results indicate a structured, automated question generation approach and found it efficient. Rathi et al. [18] explored how a system using NLP took input text, extracted important information to create questions and answers. Hang et al. [9] designed a MCQGen, an AI tool that automates the creation of personalized multiple-choice questions (MCQs) using GPT-4 and retrieval-augmented generation that adapts question difficulty based on student performance. The system produces clear, relevant questions that adjust in difficulty based on how students respond. Rathi et al. [17] explored how deep learning methods generate questions automatically and assessments utilizing NLP and transformers to create diverse question types, such as multiple-choice, short answer, and fill-in-the-blank questions. The system also integrates answer evaluation capabilities. Nwafor et al. [16] used N-grams with TF-IDF to find the keywords in the sentences and generated questions around them. Denna et al. [7] used Named Entity

Recognition (NER) on tokens to identify the terms that carried more importance in the sentence for generation MCQs along with distractors. Das et al. [5] first identified the simple sentences in the corpus. From simple sentences, key phrases were extracted using POS analysis. From key phrases, N-grams were chosen based on their Dice coefficients and frequencies, as keywords. Replacing these keywords with blanks generated Cloze Questions. This approach was optimized by using RAKE score instead of Dice coefficient and removing the need to create N-grams by Das et al [6]. Kumar et al. [14] randomly selected 2900 sentences from the book named “Operating System Principles” for analysis. The paper went an approach that was combination of ontology and machine learning. Ontology was used for generating WH-type questions and SVM was used in ML for Cloze questions. Virani et al. [23] automated the process of generating context-aware questions using Retrieval Augmented Generation (RAG). (Refer Table 2).

3 Techniques Used in Question Classification

3.1 *Revised Bloom’s Taxonomy*

Bloom’s taxonomy introduced by Benjamin Bloom in 1956 is a framework that categorizes learning objectives into six hierarchical cognitive levels, as shown in Fig. 1. These domains are used by educators to design curricula, assessments, and teaching methods to promote different types of learning. The taxonomy underwent a significant revision in 2001 by Anderson and Krathwohl. Examples and description of categories in Bloom’s taxonomy are shown in Table 3 .

3.2 *Question Collection*

A question classification model based on Bloom’s taxonomy requires a large amount of data which can be gathered from a variety of sources, including previous year question paper (PYQs), online sources, books, and previous research works datasets. The questions must be labeled based on the above-mentioned six cognitive levels.

3.3 *Pre-processing*

Questions collected from different sources can be of different formats such as text and images. This stage involves converting original questions to relevant format such as text with a proper set of attributes. Following are the steps involved [2]:

Table 2 Comparison between various research studies in generation

Author	Approach	Technique	Outcome	Gap analysis
[24]	Rule-based	Developed a model to generate Bloom's taxonomy-aligned questions using key phrase extraction and context-free grammar (CFG) rules Key steps: (1) extract NN and JJ phrases, (2) pair adjectives with nouns, (3) generate template-based questions with CFG	Contextual questions had a higher BLEU score (0.921) but a lower Cohen's Kappa (0.627) than non-contextual ones (BLEU: 0.861, Kappa: 0.76). Classification models (decision trees, random forest, gradient boosting) achieved 0.99 accuracy in relevance evaluation	– Did not explore ML and DL methods for question generation – Focused only on questions for software engineering course
[18]	ML-based	From the input text, system finds important information and creates questions with answers. These are then checked through manual reviews	73% of questions were logically and grammatically correct. 8% had logical questions but inaccurate answers. 19% were illogical or irrelevant to the text	Limited question types
[9]	DL-based	MCQGen is an AI tool that automates personalized MCQ creation using GPT-4, RAGs, and adaptive difficulty based on student performance, aiming to address various learning needs and personalize education	MCQGen creates clear, relevant questions that adjust in difficulty based on student answers	– Subject-specific focus (only computer science) – Exclusive use of GPT-4 without model comparison
[17]	DL-based	The study focuses on NLP and transforms to create diverse question types, including multiple-choice, short answer, and fill-in-the-blank questions, and also performs answer evaluation	T5 and BERT-based fine-tuned models performed best on SQUAD dataset with scores as T5 (F1:63.28, EM:37.24) and BERT (F1: 77.19, EM:57.81)	– Did not focus on personalizing the generated questions – Did not explore hybrid models

(continued)

Table 2 (continued)

Author	Approach	Technique	Outcome	Gap analysis
[16]	ML-based	A system for automated MCQs generation using NLP techniques. The system extracts keywords from lecture materials to create relevant questions. It utilizes methods like TF-IDF and N-grams to identify significant words and generates MCQs based on these keywords	Recall is high (0.84-0.98), most important keywords were successfully extracted. Precision is moderate (0.25-0.44), some irrelevant keywords were also extracted. F1-measure is balanced (0.39-0.58), effective but with potential for improvement	Generated only fact-based questions
[7]	Rule-based	A system was developed to automatically generate MCQs using key terms from the content, with correct answers and distractors, and subjective questions based on Bloom's taxonomy to test various cognitive levels	84% precision, 73% recall in QG evaluated by 20 teachers, questions matched manual quality. 70 MCQs generated from 80 sentences with 59 relevant	Stored all the generated questions without considering user's response to them
[5]	Rule-based	Used Stanford Parser, CoreNLP, and typed dependency manual to identify simple sentences and extract key ones with rule-based POS analysis. Formed two groups: G1 (trigrams based on proper noun co-occurrence via Dice coefficient) and G2 (unigrams). Replaced blanks with the top Dice coefficient trigram or most frequent unigram	Achieved 92.367% for informative simple sentence identification when judged by 5 linguistic experts	Generated only the questions of fill in the blanks type

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Table 2 (continued)

Author	Approach	Technique	Outcome	Gap analysis
[6]	Rule-based	Converted the corpus into simple sentences using methods from “A Novel System for Generating Simple Sentences.” Used RAKE to extract keywords (NNP, NNPS, CD), selected the highest-scoring keyword as the stem and replaced it with a Wh-word based on its category from Stanford Named Entity Recognizer	The time required by the system to execute increased with the count of sentences in the dataset. The informative sentences that are ranked in the top are found to be more suitable to be used as stem of MCQ generation because they have more informative keywords	– Limited Dataset – Did not explore ML- and DL-based methods for automating the question generation
[14]	ML-based	Proposed a hybrid approach for MCQ generation with an ontology for Wh-type questions and ML for Cloze questions. Evaluated MCQs using Bloom’s taxonomy and tested the framework on an operating systems dataset	The questions of cloze type are also found to be useful in applications that are real-time. The generated questions satisfied the cognitive skills of level 1 & 2 which are based on Bloom’s taxonomy	Generated questions of Cloze and Wh-type and hence didn’t explore MCQ generation
[23]	DL-based	Used Hugging Face Transformers library to load a pre-trained T5 model and generate Factoid Wh-Type QA, MCQ-based QAG, Full Sentence QAG, Gap-Fill QAG, and Boolean QAG	Accuracy of the QA was evaluated by comparing with original text and verifying the answers manually	The accuracy of generated distractors could be improved, and the generated questions were evaluated manually

1. **Tokenization:** It breaks the text into parts known as tokens (words or sentences or punctuations) based on the methods used for tokenization. It is required to make the text manageable for further steps.
2. **Stopword removal:** Here, stopwords, i.e., the words that don’t have meaning individually, are removed as they don’t contribute to the context of the text.
3. **Tagging:** Tagging is the procedure by which the tokens are attached with the part of speech they belong to. This step helps the model understand the meaning of words in the sentence as words when used as different parts of speech have different meanings.

Table 3 Examples of categories in revised Bloom's taxonomy

Category	Description	Example
Remember	Recalling facts of which we are already aware about	Define mean
Understand	Putting together the meaning of facts and ideas by organizing and summarizing them. It also involves activities such as interpreting and inferring	Explain the difference between mean and median
Apply	Using the knowledge which is gained to solve problem in new situation	Calculate the average height of students in your class
Analyze	Dividing the information into smaller parts to study relations between them	Analyze the relationship between the radius and the circumference of a circle
Evaluate	Making conclusions based on standards or criteria	Evaluate the steps taken in solving the problem $3 \times +4=19$. Identify and correct any mistakes
Create	It involves merging the elements to create a meaningful functional final product	Create a word problem involving a system of equations, and provide a solution

4. **Stemming:** Stemming changes the POS tagged tokens to their root forms. This step is required to bring all the forms of word under one umbrella to be handled as one and not different tokens.
5. **Parsing:** Parsing helps understand the semantics of the tokens by building the parse tree. To understand grammatical errors in the text.

3.4 Feature Extraction

Here raw input data is transformed into useful set of features that can be interpreted by machine learning classifiers [19].

- **Lexical:** Lexical features capture how words are related within a question. Using n-grams allows us to capture not only individual words but also the context in which they appear [13].
- **Syntactic:** Relates to the structure and grammatical patterns in a question [13].
- **Semantic:** It focuses on the meaning of words and their relationships in the context of a question [13].

- **Principal Component Analysis (PCA)** [20]: It is a dimensionality reduction technique that transforms the original features into a new set of uncorrelated features called principal components. These components are linear combinations of the original features and are ranked by the amount of variance they capture in the data.
- **Latent Semantic Index (LSI)** [20]: Another dimensionality reduction technique of NLP, which applies Singular Value Decomposition (SVD) to a term-document matrix, transforming the text data into a lower-dimensional semantic space.

3.5 Term Weighing

Binary method represents the simplest approach to text encoding where each word or phrase in a question is converted into a simple vector of 0s (indicates that a term is absent) and 1s (indicate its present). To address the limitations of the binary method, **Term Frequency (TF)** introduces a measure of significance based on the recurrence of terms within a question. TF assigns higher values to terms that appear frequently within a given query, reflecting their potential relevance.

$$TF_{t,d} = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d} \quad (1)$$

Although TF provides insight into term significance within a question, it has limitations, particularly in handling common words that appear frequently in many questions (e.g., “a,” “an”) but may carry limited informational value. **TF-IDF** addresses these limitations by combining Term Frequency (TF) Inverse Document Frequency (IDF), which decreases the importance of terms that are widespread across the entire dataset.

$$TF_{t,d} = \frac{f_{t,d}}{N_d}$$

where

- $f_{t,d}$: Frequency of term t in document d .
- N_d : Total number of terms in document d .

$$IDF_t = \log \frac{N}{n_t}$$

where

- N : Total number of documents in the corpus.
- n_t : Number of documents containing the term t .

$$\text{TF-IDF}_{t,d} = \left(\frac{f_{t,d}}{N_d} \right) \times \log \frac{N}{n_t} \quad (2)$$

The **Enhanced TF-IDF (E-TFIDF)** further refines the TF-IDF model by incorporating class distribution and part-of-speech weighting into term weighting. In this approach, terms are given different weights depending on their part-of-speech (POS) classification like verbs, nouns, and adjective and how they are distributed across different classes. This is because certain types of words like verbs or nouns can hold varying levels of importance in understanding the context and meaning of a question. By assigning distinct weights to different POS categories, the model can more accurately capture the relevance of each term in question classification.

$$\text{Enhanced TF-IDF}_{t,d} = \left(\frac{f_{t,d}}{N_d} \times L_d \right) \times \log \frac{N}{n_t + 1} \quad (3)$$

where

- $f_{t,d}$: Frequency of term t in document d .
- N_d : Total number of terms in document d .
- L_d : Normalization factor for document d , e.g., maximum term frequency or adjusted length.
- N : Total number of documents in the corpus.
- n_t : Number of documents containing the term t (with +1 to avoid division by zero).

Term Frequency Inverse Document Frequency (TFPOS-IDF) is a further evolution of TF-IDF and E-TFIDF emphasizing POS roles in text classification. Terms are assigned specific weights based on their POS category with the aim of capturing the POS in query classification.

$$\text{TFPOS-IDF}_{t,d} = \left(\frac{f_{t,d} \cdot w_{\text{POS}_t}}{N_d} \right) \times \log \frac{N}{n_t + 1} \quad (4)$$

where

- $f_{t,d}$: Frequency of term t in document d .
- w_{POS_t} : Weighting factor based on the part of speech (POS) of term t , adjusted to reflect its grammatical role, such as noun, verb, or adjective.
- N_d : Total number of terms in document d .
- N : Total number of documents in the corpus.
- n_t : Number of documents containing the term t (with +1 to avoid division by zero).

RoBERTa is a modified version of BERT that uses fewer parameters and omits the next sentence prediction task. It has undergone longer training periods and utilized a larger dataset compared to BERT, enhancing its performance [8].

3.6 Feature Selection

It involves selecting those features that are more important than others for categorization.

- **Document Frequency (DF)** [20]: It refers to the number of documents in a collection that contain a particular term, and if the document frequency of a particular term is less than some threshold, then that term will be eliminated.
- **Information Gain (IG)** : It tells us how much a feature contributes to making better predictions. A higher information gain indicates that the feature is more useful for predicting the target variable. IG is given by the following equation

$$\begin{aligned} \text{IG}(t) = & - \sum_{i=1}^M p(C_i) \log p(C_i) + p(t) \sum_{i=1}^M p\left(\frac{C_i}{t}\right) \log p\left(\frac{C_i}{t}\right) \\ & + p(\bar{t}) \sum_{i=1}^M p\left(\frac{C_i}{\bar{t}}\right) \log p\left(\frac{C_i}{\bar{t}}\right) \end{aligned} \quad (5)$$

- **Gini Index (GI)**: The Gini Index is a measure of how well a split separates the classes at a node in a decision tree. It is widely used for selecting the best splits due to its simplicity, helping create trees that separate classes with minimal impurity.

$$\text{Gini}(D) = 1 - \sum_{i=1}^n p_i^2 \quad (6)$$

where

- D : The dataset or subset of data being evaluated.
- n : The total number of classes.
- p_i : Probability of selecting a data point in class i .
- **Chi-square**: It is used to determine if there is a significant association between two categorical variables.

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (7)$$

where

- O_i : Observed frequency of term i in a class.
- E_i : Expected frequency of term i in a class, assuming independence.
- n : Number of classes or categories.
- **Mutual Information (MI)**: It quantifies the mutual dependence between two random variables.

$$MI(t, c) = \sum_{t \in \{0,1\}} \sum_{c \in \{0,1\}} p(t, c) \log \frac{p(t, c)}{p(t) \cdot p(c)} \quad (8)$$

where

- $p(t, c)$: Joint probability of term t and class c .
- $p(t)$: Probability of term t occurring.
- $p(c)$: Probability of class c occurring.

- **Odds Ratio (OR)** : It is a measure of association between an exposure and an outcome.

$$OR(t, c) = \frac{P(t|c)/(1 - P(t|c))}{P(t|\neg c)/(1 - P(t|\neg c))} \quad (9)$$

where

- $P(t|c)$: Probability of term t given class c .
- $P(t|\neg c)$: Probability of term t given the complement of class c (not c).

Other techniques are Best Term and Ambiguity Measure (AM).

3.7 Classification

Classification is supervised learning technique that learns to categorize data into given class labels. Following are the classification algorithms used in the articles reviewed:

- **Linear Discriminant Analysis** [11]: It measures the separation in classes created by linear combination of features. Then selects the features that create maximum separation in classes. It is hence also used in feature extraction to select top k features that divide the data more clearly in given classes.
- **Logistic Regression** [11]: It is used to predict the likelihood of an instance belonging to a specific class. This is used in binary classification where a sigmoid function takes independent variables as input and outputs a probability between 0 and 1. To use logistic regression for multiclass classification, one-vs-rest strategy is used. For example, six binary classification tasks are considered to determine the question class, such as identifying whether the class is comprehension or knowledge, and so on with all classes [19]. The predicted class is then determined using Maximum Likelihood Estimation, which is calculated using the following equation:

$$L(\beta) = \prod_{i=1}^m P(y_i|x_i, \beta)^{y_i} \cdot (1 - P(y_i|x_i, \beta))^{1-y_i} \quad (10)$$

$$\log L(\beta) = \sum_{i=1}^m (y_i \log(P(y_i|x_i, \beta)) + (1 - y_i) \log(1 - P(y_i|x_i, \beta))) \quad (11)$$

where

- m : Total number of samples.
 - y_i : Observed label for the i -th sample (0 or 1).
 - $P(y_i|x_i, \beta)$: Predicted probability of y_i given x_i under parameters β .
 - β : Parameters (coefficients) of the logistic regression model.
- **Naive Bayes** [1, 3]: The Naive Bayes classifier does probabilistic classification which is derived from Bayes theorem. During training, it calculates the conditional probabilities between features and target variables. Using them, it gives the probabilities for a given set of features to belong to each of the classes. It assumes the features are independent of each other. It is useful to model problems like weather prediction, etc.
 - **Decision Trees** [11]: Decision trees model decisions based on a series of questions that lead to a conclusion. It starts at the root node and evaluates the feature that best splits the data into different classes based on a criterion, and the process is recursively applied to each subset, creating branches until a stopping condition is met. After the tree is built, new data points can be classified by tracing a path from the root to a leaf node, making decisions at each step based on the feature values.
 - **K-NN** [1, 11]: K-nearest neighbor is a graphical algorithm that plots the data as points in multi-dimensional space, where dimensions are equal to features. When new data points are introduced for prediction, its distance is calculated from each of these data points using various distance metrics like Euclidean, Hamiltonian, etc. Then closest k data points, known as neighbors are chosen. Each of these neighbors contributes 1 vote for their class, hence using the voting algorithm; the class with most votes is then predicted as the class for the given data point. This is useful to model low-dimensional and nonlinear data.
 - **SVM** [1, 11, 21, 25]: A support vector machine (SVM) tries to find the best line or boundary (hyperplane) which separates different classes in the feature space. SVMs work especially well with high-dimensional data, thanks to kernel functions that help create this separation and are less likely to overfit. The linear kernel equation is: SVM Decision Function:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \quad (12)$$

Decision Rule:

$$\text{Class} = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x} + b \geq 0 \\ -1 & \text{if } \mathbf{w}^T \mathbf{x} + b < 0 \end{cases}$$

where

- x is the input feature vector.
 - w is the weight vector (normal vector to the hyperplane).
 - b is the bias term (offset from the origin).
 - $w^T x$ represents the dot product between the weight vector and the feature vector.
- **CNN [15]**: Convolutional neural networks are efficient in modeling the linear dependencies because they process data sequentially using a process known as convolution. For text analysis, 1D CNN is used. It calculates the dot product between the filter and group of words in text. Each filter captures a certain pattern in text. Hence, it is useful for problems like movie genre classification, etc., where linear dependency affects the classification.
 - **Random forest [11]**: A method that merges several decision trees to boost classification accuracy and make predictions more dependable. Also it helps to reduce the risk of overfitting that single decision tree might experience. It uses techniques such as bagging, where each tree is trained on a random subset of the original dataset and random feature selection. It is a flexible method for multiclass classification tasks.
 - **Neural Network [11]**: Neural networks are powerful for classification tasks as they naturally capture nonlinear relationships without requiring manual feature selection. They outperform traditional machine learning methods by assigning weights to features based on their importance in determining the final output. These networks are made up of layers: an input layer that receives data, an output layer that provides predictions, and one or more hidden layers responsible for learning patterns. Each neuron in a layer connects to neurons in the next layer, with each connection having weights and biases that adjust as the network learns. Training involves passing input through the network, computing an error, and updating weights to minimize it. This process continues until the network learns optimal values. During prediction, inputs move through layers, getting modified by activation functions, which introduce nonlinearity. The final layer assigns probabilities to each class, and the one with the highest probability is chosen as the predicted label.

3.8 Evaluation Metrics

To assess the performance of a model, metrics such as accuracy, precision, recall, and F1-measure are calculated. These metrics are defined using the concepts of True Positives (TPs), False Positives (FPs) and False Negatives (FNs) as shown in Fig. 2 [8, 19] True Positives represent the instances that are incorrectly classified correctly and identified as belonging to the positive class, while False Positives are those which are incorrectly classified as positive and False Negatives are instances that belong to the positive class but are not misclassified as negative. Recall, accuracy, F1-measures

Fig. 2 Evaluation metrics

		ACTUAL	
		Positive	Negative
PREDICTED	Positive	True Positive(TP)	False Positive(FP)
	Negative	False Negative(FN)	True Negative(TN)

are scaled between 0 and 1. The values nearer to 0 signify poor performance, and values closer to 1 signify strong performance.

- **Precision:** Checks how accurate the model's positive predictions are by measuring the proportion of true positives out of total positive predictions

$$\text{Precision} = \frac{TP}{TP + FP} \quad (13)$$

- **Recall:** Measures the model's effectiveness to capture all relevant positive cases, calculated as the ratio of true positives out of total actual cases.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (14)$$

- **F1-Measure:** Balances precision and recall to get a single score, especially useful if one of the metrics outweighs the other.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{TP}{2TP + FP + FN} \quad (15)$$

4 Methods Used in Question Generation

4.1 Rule-Based Methods

- **Named Entity Recognition (NER)** is used to create MCQs by pulling out important terms like names, organizations, and places from the text. This enables the system to produce correct answers and distractors for each question. Deena et al. [7] NER helps to determine the category of the selected answer key from the informative sentence by recognizing whether the answer refers to a noun and the system

can replace it with the suitable “wh-word” in the question stem. Das et al. [6]. This ensures that the questions are not only relevant but also clear and engaging the learners.

- **Context-Free Grammar (CFG)** contains grammar rules written recursively, used to build strings that follow the given pattern. If the question is structured by combining adjective phrases with Bloom’s taxonomy verb, CFG generates templates that can be used for generating questions that follow Bloom’s taxonomy [24].

4.2 Machine Learning-Based Methods

- **Using POS tags:** Words with POS tags of nouns in key phrases become the keywords for that sentence. The sentence when these keywords are replaced with blanks becomes Cloze Questions [16, 18].
- **SVM :** SVM classifies the sentences as key phrases. Then the keywords chosen from these keyphrases when replaced with blanks generate Cloze questions [14].

4.3 Deep Learning-Based Methods

- **T5** is built by Google as text-to-text transformer. It was built to create a unified model that could solve all ML problems without need to change the architecture. Hence, it is pre-trained on different tasks by converting all of them to text-to-text format. Fine-tuning T5 with custom text and questions based on that text creates a model that could generate questions from given text [17, 23].
- **GPT-4** Generative Pre-trained Transformer 4 is 4th in the series of GPTs released by OpenAI and brings multimodal capability with it. It is trained on large data with a lot of variety in domains; hence, it has broad understanding in various domains. With its transformer architecture, it is able to capture semantics of large sequences of data. This model of GPT when used by instructors generates semantically correct and domain-related questions [9].
- **RAG** Retrieval Augmented Generation is an enhancement made to LLMs, where instead of directly answering questions; the relevant information is queried from vector databases and is synthesized to create a knowledge base before answering the question. The performance of LLMs like GPT-4 can be improved when attached to external database like the questions prepared by teachers and students, thus incorporating RAG [9].

4.3.1 Evaluation Metrics

The evaluation of Automated Question Generation (AQG) techniques is important for assessing the effectiveness of quality of questions generated. To evaluate how well an

AQG system performs two key metrics are often used: precision and the F-measure. Precision looks at the accuracy of the system by showing how many of the generated questions are actually relevant. Das et al. [5] The F-measure balances precision with recall which measures how many relevant questions the system generates out of all possible relevant ones. That helps in assessing the system's overall performance when there is a need to balance accuracy with coverage [16]. Together, these metrics help to understand how well the AQG system can generate questions that are accurate and relevant both. Then teachers reviewed the questions to confirm they were clear and relevant [16]. The correctness of the generated questions and answers was checked by matching them with the original text and manually confirming the answers. Das et al. [23]. For the evaluation method used by [5], manual evaluation was conducted by linguistic experts to assess the accuracy of sentence selection and answer-key identification was manually verified for correctness. Another metric for evaluation is the Bilingual Evaluation Understudy (BLEU) score that measures the machine translations by comparing the n-grams of machine translated sentences with the n-grams of human-translated sentences. Rathi et al. [17] used BLEU score to score the distractors.

5 Conclusion

This review has highlighted the importance of question classification and automatic question generation (AQG) in modern educational assessment, based on Bloom's taxonomy for evaluating cognitive development. The study highlights that hybrid machine learning approaches often outperform traditional rule-based methods in classification tasks, provided that sufficient high-quality training data is available. Current research in question classification highlights the need for further exploration of deep learning approaches, as existing methods primarily rely on traditional techniques. To address this gap, our future research will focus on consolidating advanced deep learning models to enhance classification accuracy and adaptability. Similarly, in the domain of AQG, RAG models remain underexplored. Given their potential for generating contextually relevant and high-quality questions, we aim to investigate their effectiveness in AQG to improve question diversity and coherence. Looking ahead, our ultimate goal is to develop a unified personalized platform that seamlessly integrates both classification and question generation. This system will enable students to practice exam questions, receive automated evaluations, and enhance their learning experience through AI-driven feedback. By bridging these research gaps, we envision a more effective and intelligent educational tool that supports learners.

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